AVANISH TIWARI

Senior Frontend & Fullstack Engineer (React, TypeScript, Node.js)

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PROFESSIONAL SUMMARY

Frontend & Fullstack Software Engineer with 5+ years of experience engineering complex client-side logic systems, reusable component architectures, and responsive web applications across the full SDLC. Specializes in React, Next.js, TypeScript, and Node.js with a proven track record in web performance optimization (code-splitting, microfrontends, and load time reductions up to 60%). Experienced in designing agentic AI pipelines, local browser model runtimes (WebGPU, WASM), and provisioning cloud hosting infrastructure on AWS (EC2, Elastic IP, R2). Strong background in REST APIs, relational databases (PostgreSQL, Prisma), and agile delivery.

TECHNICAL SKILLS

- **Frontend & Core:** React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Vite, Webpack, Babel
- **Performance & Architecture:** Core Web Vitals, SSR/SSG/CSR, Hydration Fundamentals, Code-Splitting, Microfrontends, State Management
- **Backend & Databases:** Node.js, Express.js, REST APIs, PostgreSQL, Prisma ORM, JWT, bcrypt
- **Cloud & Infrastructure:** AWS (EC2, Elastic IP), Cloudflare (R2, DNS), Linux, Nginx/Reverse Proxy, Custom Domain Management, Vercel, Render
- **AI Integration & Runtimes:** Function Calling / Tool Use, Model Context Protocol (MCP), LLM APIs (Groq, Gemini), Browser Runtimes (Chrome AI, WebGPU, WASM)
- **Testing & Methodologies:** Jest, Git, GitHub, Code Reviews, Agile (Jira), MVC Architecture

PROFESSIONAL EXPERIENCE

Senior Scripting Executive (Frontend & Web Applications) — Ipsos *May 2025 – Present | Noida, India*

- Architect and maintain complex client-side logic, dynamic state branching, and validation rules for high-volume enterprise survey web applications.
- Build modular, accessible question-block templates and reusable React-based UI components, enabling non-technical teams to assemble applications faster.
- Boosted UI rendering performance by 25% and build efficiency by 40% by contributing to a scalable microfrontend architecture.
- Document component specifications and design patterns to ensure maintainability and seamless adoption across development teams.
- Architected an internal multi-agent automation workflow (Review, Structure, Logic, and Audit agents) to generate survey logic and schema directly from source specs, enforcing strict guardrails to prevent regressions.
- Collaborate cross-functionally with senior stakeholders and engineers on API integrations, debugging, and feature rollouts.

Scripting Executive (Frontend & Web Applications) — Ipsos *Oct 2022 – Apr 2025 | Noida, India*

- Developed and tested dynamic logic systems and input validation rules for responsive web applications, reducing rework from edge-case logic failures.
- Designed and documented reusable application templates and UI modules, cutting survey build turnaround time by 30%.
- Conducted regular peer code reviews and enforced standard coding practices to improve code quality and long-term maintainability.

Frontend Developer — Phronesis Partners *Nov 2021 – Oct 2022 | Noida, India*

- Modernized web application templates using code-splitting, lazy loading, and asset optimization, improving load times by 60% and enhancing Core Web Vitals.
- Integrated RESTful APIs with frontend client state to deliver responsive, data-driven interfaces.
- Collaborated with UI/UX designers to implement pixel-perfect, responsive web interfaces across devices.

Product Trainee — Ameyo *Feb 2021 – Oct 2021 | Gurugram, India*

- Assisted in backend and product feature development using Linux and PostgreSQL, supporting IT and integration requirements.

KEY TECHNICAL PROJECTS

Personal Web Platform & Infrastructure (avanishtiwari.in) *AWS EC2, Elastic IP, Cloudflare R2, Linux, Nginx, Custom Domain DNS*

- Configured and deployed personal domain and portfolio web application on a cloud-hosted AWS EC2 Linux instance.
- Provisioned fixed Elastic IP addressing, configured Nginx reverse proxy, and secured routing with SSL/TLS certificates.
- Integrated Cloudflare R2 storage for optimized static asset rendering, caching, and low-latency global delivery.

Private AI Shield *React, Chrome AI (Gemini Nano), WebGPU, WASM*

- Built an offline-first, privacy-focused browser tool running client-side AI text processing without external API calls or backend servers.
- Implemented a tiered execution engine (Chrome AI → WebGPU → WASM → Fallback) with automatic graceful degradation.
- Developed real-time PII detection and live UI warning banners to alert users before submitting sensitive data.

Autonomous Survey Engineering Multi-Agent System *TypeScript, Node.js*

- Designed a 4-stage pipeline (Review → Structure → Behavior → Audit) separating declarative schema authoring from runtime logic execution.
- Created deterministic validation scripts to verify string escaping, conditional branching, and categorical logic against client source documents.

AI TaskFlow (Live Link) *React, Vite, Node.js, Express.js, PostgreSQL, Prisma, Groq API*

- Fullstack task management application integrating Groq AI (Llama 3.3 70B) through an Express backend service.
- Managed relational PostgreSQL data models with Prisma ORM; resolved CORS and environment configuration for production.

Job Application Tracker (Live Link) *React, Node.js, Express.js, PostgreSQL, JWT, bcrypt*

- Designed relational database schema with cascading deletes and built full CRUD REST endpoints.
- Implemented JWT authentication and secure Express middleware for route authorization.

EDUCATION & CERTIFICATIONS

- **B.Tech in Computer Science:** United College of Engineering and Research, AKTU (2016 – 2020)
- **Certifications:** Back-End Development (IBM SkillsBuild, Mar 2026) | React (Scaler) | JavaScript Algorithms (freeCodeCamp) | Node.js Essential Training (LinkedIn)